Project Title:

Factors Affecting Student Academic Performance: A Statistical Analysis Using Portuguese Secondary School Data in R

Introduction:

The objective of this project/study is to find the factors that affect the students' academic performance using multiple linear regression model. The analysis is conducted in R where a lot of explanatory variables were examined.

Data Source:

The dataset was introduced in the paper:

Student Performance Data Set by **Paulo Cortez** and **Alice Maria Gonçalves Silva**.

Original Data repository:

https://archive.ics.uci.edu/dataset/320/student+performance

The data used in this study was obtained from the UCI Machine Learning Repository. The dataset contains demographic, social, family, and academic information for Portuguese & math's secondary school students. It has been widely used for educational data mining and predictive modelling.

Variables:

Response Variable:

Average grade is the response variable in this study.

Average grade (average of mathematics and Portuguese final grades)

Response variable was created by taking average of these two variables:

G3_maths: student's final grade in subject Mathematics.

G3_port: student's final grade in subject Portuguese.

Student scoring high in final grades is likely to have higher average grade.

Explanatory variables:

Initially, all of the predictors were included in the study excluding below variables:

- G1 - first period grade (numeric: from 0 to 20)
- G2 - second period grade (numeric: from 0 to 20)
- G3 - final grade (numeric: from 0 to 20, output target)

These variables were excluded because they were highly correlated with our response variable.

Initially, all of the predictors were included in the model but there are more than 25 predictors and explaining all of them can take a lot of time to understand. So, I am going to explain only those explanatory variables that were included in the final model.

school - student's school (binary: "GP" - Gabriel Pereira or "MS" - Mousinho da Silveira)

address - student's home address type (binary: "U" - urban or "R" - rural)

famsize - family size (binary: "LE3" - less or equal to 3 or "GT3" - greater than 3)

Medu - mother's education (numeric: 0 - none, 1 - primary education (4th grade), 2 – 5th to 9th grade, 3 – secondary education or 4 – higher education)

Mjob - mother's job (nominal: "teacher", "health" care related, civil "services" (e.g. administrative or police), "at_home" or "other")

traveltime - home to school travel time (numeric: 1 - <15 min., 2 - 15 to 30 min., 3 - 30 min. to 1 hour, or 4 - >1 hour)

failures - number of past class failures (numeric: n if 1<=n<3, else 4)

famsup - family educational support (binary: yes or no)

higher - wants to take higher education (binary: yes or no)

goout - going out with friends (numeric: from 1 - very low to 5 - very high)

health - current health status (numeric: from 1 - very bad to 5 - very good)

studytime - weekly study time (numeric: 1 - <2 hours, 2 - 2 to 5 hours, 3 - 5 to 10 hours, or 4 - >10 hours)

schoolsup - extra educational support (binary: yes or no)

romantic - with a romantic relationship (binary: yes or no)

absences - number of school absences (numeric: from 0 to 93)

Data Preparation:

- First, we imported both datasets file (maths data and Portuguese data) and merged them based on the observations that match in both datasets (inner join) together in a single data named students.
- Converted categorical variables into factor by using R.
- Checked if there were any missing values, none found.
- Checked if there were any duplicate values, none found.
- Created a function to check number of categories of any desired variable

Exploratory Data Analysis:

- Checked descriptive statistics of variables.
- Checked outliers by visualizing boxplot of a variable. Decided to keep outliers in variables because they were actually reasonable value in a variable.
- Observed the distribution of response variable by constructing histogram. The distribution was found to be normal.
- Made bar plots for some categorical variables and also defined a function to simply use it to create a bar plot for any specific variable.
- Heatmap plot was created to observe correlations between the variables.

Model Building:

Initially, all of the explanatory variables were included in the model but after seeing the non-significant results, I decided to run a backward elimination method of multiple regression to find the best model which finds the best model based on AIC criterion in R language through the function ‘step()’ as:

full_mdl = lm(average_grade ~ ., data = students)

backward_model = step(full_mdl, direction = "backward")

summary(backward_model)

Result of model:

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	**10.4178951**	**1.20286504**	**8.66090107**	**1.5587E-16**
schoolMS	**-1.0998945**	**0.50775823**	**-2.1661777**	**0.03094797**
addressU	0.57475196	0.38524845	1.49189946	0.13659403
famsizeLE3	0.58556564	0.32604193	1.79598261	0.07332874
Medu	0.30748283	0.1796653	1.71142021	0.08785777
Mjobhealth	0.89353778	0.71810857	1.24429344	0.2141944
Mjobother	-0.1427428	0.46677226	-0.3058081	0.75992603
Mjobservices	0.81860117	0.52804498	1.55024895	0.12195315
Mjobteacher	-0.1702476	0.65630353	-0.2594038	0.79547077
traveltime	-0.1272264	0.22634448	-0.5620918	0.57440036
failures	**-1.3900165**	**0.22177468**	**-6.2676971**	**1.0395E-09**
famsupyes	-0.2733915	0.30681966	-0.8910495	0.3734928
higheryes	**1.99693123**	**0.73604267**	**2.71306449**	**0.00698403**
Goout	**-0.2966354**	**0.12848302**	**-2.3087519**	**0.02151849**
health	**-0.2250408**	**0.10445127**	**-2.1545052**	**0.031858**
Studytime	0.30583908	0.17869868	1.71147925	0.08784687
Schoolsupyes	**-1.4853092**	**0.43358946**	**-3.4256118**	**0.00068361**
romanticyes	-0.5698313	0.3119119	-1.8268984	0.06853606
absences	-0.0523631	0.03055594	-1.7136787	0.08744146

The values are bold whose p-value < 0.05 in the above table.

It keeps those non-significant variables as well by which AIC is low. So, our final model contains some non-significant variables (p-value > 0.05) as well.

Equation of the final model:

Average grade = **10.41** + **(-1.09 * school$_{\textbf{Mousinho da Silveira}}$)** + (0.574 * urban area) + (0.585 * family size$_{<=3}$) + (0.307 * Mother’s education) + (0.893 * Mother’s job $_{\text{health}}$) + (-0.142* Mother’s job $_{\text{other}}$) + (0.818 * Mother’s job $_{\text{services}}$) + (- 0.170* Mother’s job $_{\text{teacher}}$) + (-0.127 * traveltime) + **(-1.39 * failures)** + (-0.273 * family_support$_{\text{yes}}$) + **(1.996 * higher education$_{\textbf{yes}}$)** + **(-0.296 * Going out with friends)** + **(-0.225 * current health status)** + (0.305 * weekly study time) + **(-1.48 * extra educational support$_{\textbf{yes}}$)** + (-0.569 * in a relationship) + (-0.052 * number of school absences)

In the above model, I have made variables bold that are statistically significant (p-value < 0.05) and there are only 6 predictors that are statistically significant. Let’s explain each of the significant variables first.

Significant predictors:

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	10.4178951	1.20286504	8.66090107	1.5587E-16
schoolMS	-1.0998945	0.50775823	-2.1661777	0.03094797
failures	-1.3900165	0.22177468	-6.2676971	1.0395E-09
Higheryes	1.99693123	0.73604267	2.71306449	0.00698403
Goout	-0.2966354	0.12848302	-2.3087519	0.02151849
Health	-0.2250408	0.10445127	-2.1545052	0.031858
schoolsupyes	-1.4853092	0.43358946	-3.4256118	0.00068361

School: The predicted (mean of) average grade of a student that is enrolled in the Mousinho da Silveira school and studying there is estimated to be **1.09 points** lower than the student studying in the school Gabriel Pereira while holding other variables constant. This result contributes significantly in the model and it's effect is **statistically significant** ($p < 0.05$).

Failures: Holding all other variables constant, the predicted average grade decreases by **1.39 points** for each additional past class failure. This predictor is **statistically significant** at the 5% significance level (**$p < 0.05$**).

Higher education: While holding other variables constant, the predicted average grade for a student that wants to pursue higher education in future, is predicted to be **1.99 points** higher than the student that doesn't plan to pursue higher education. This predictor is found statistically significant at the 5% significance level **($p < 0.05$).**

Going out with friends: While holding other variables constant, the predicted average grade for a student is estimated to decrease by **0.296 points** for each additional unit increase in the variable **going out with friends.** This predictor is statistically significant and it has significant effect on the response variable ($p < 0.05$). Since, this predictor is ordinal variable so it means the student is likely to have a low average grade that goes out with friends a lot (example goout = 5) and on the other hand, the student is likely to perform better in academics and estimated to have higher academic grades that doesn't go out much with friends.

Current health status: Holding all other variables constant, a one-unit increase in the current health status score is associated with a 0.225-point decrease in the predicted average grade. Although this association is statistically significant, the effect size is relatively small. This finding should be interpreted as an association rather than a causal relationship and may reflect the influence of other behavioral or lifestyle factors not fully captured in the model.

Extra educational support: Holding other variables constant, the predicted average grade for a student that has extra educational support is predicted to be 1.48 points lower than the student that doesn't have extra educational support. This predictor is statistically significant ($p<0.05$).

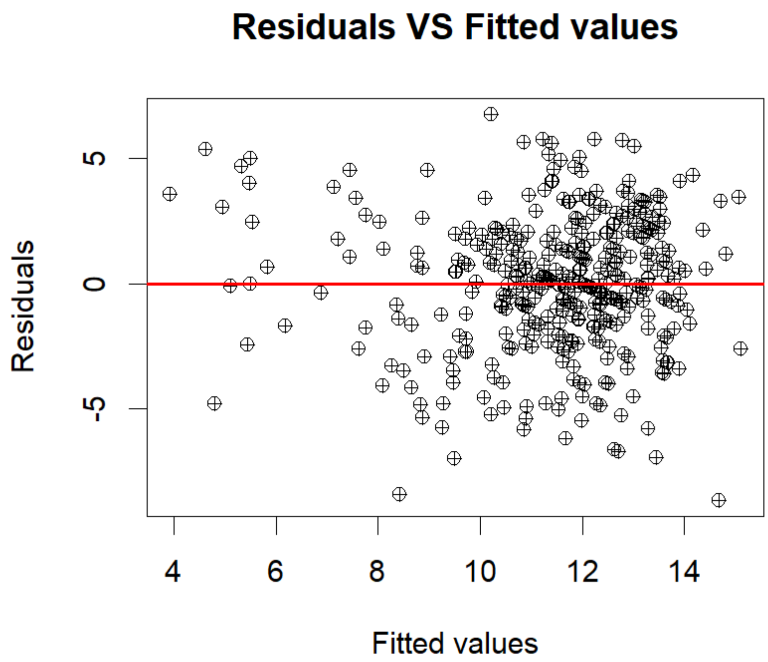
The rest of the variables are not statistically significant (p-value > 0.05) so I decided not to include and to discuss them separately. However, to take a brief summary, we can say that student's mother job, student's residence area, student's family size, support from family, student either in a relationship or not, these variables/factors do not have any statistically significant effect on the average grade of a student according to our model.

Regression Diagnostics:

The assumptions of final model were checked and none of the assumptions were found being violated.

Multicollinearity: There are no predictors that were highly correlated with each other and each predictor's VIF is around 1 which is okay. It was checked by running the command vif(final_model). So, multicollinearity assumption isn't violated.

Independence of error terms:



By examining the above plot, we can see that the points are scattered randomly around the line at y=0, which indicates that error terms are independent.

Homoscedasticity: With reference to the above plot, we can also conclude that error terms have constant variance because they don't follow any kind of pattern and points are just scattered randomly. We also conducted Breusch–Pagan test to get significant evidence of homoscedasticity, and we failed to reject the null hypothesis and conclude that error terms have constant variance. (p-value > 0.05). The results that were obtained are:

studentized Breusch-Pagan test

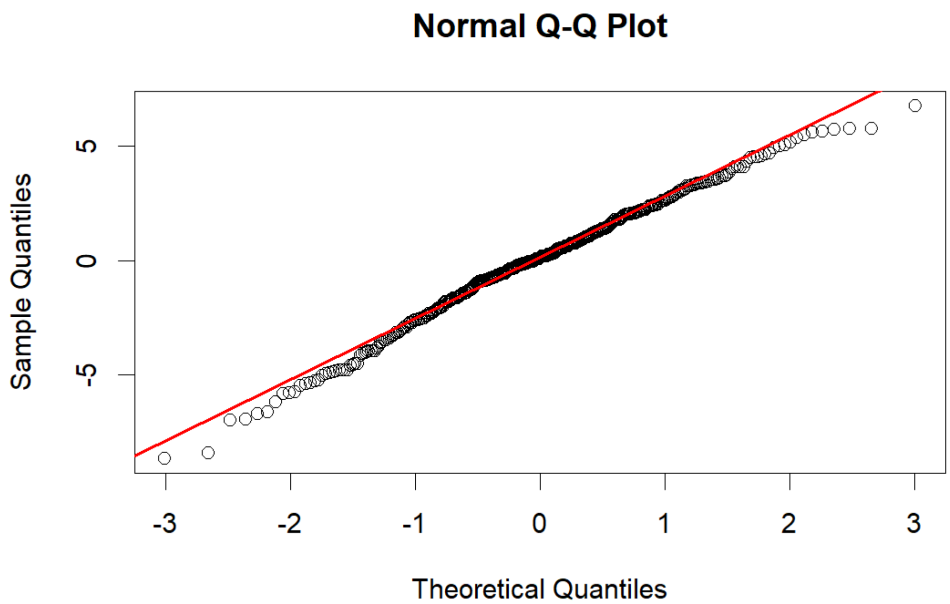
data: final_model

BP = 21.698, df = 18, p-value = 0.2457

Normality: We made a qq-plot to check if the distribution of the error terms is normal by using the command

qqnorm(resid)

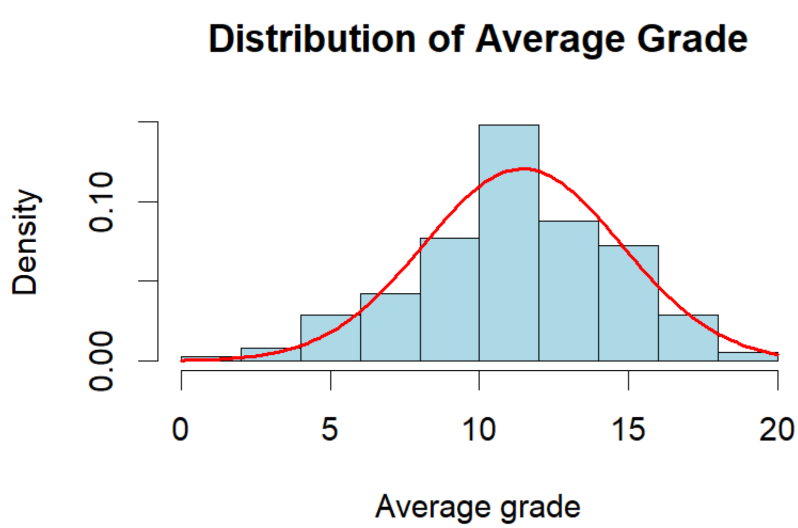
qqline(resid, col = "red", lwd = 2)



From the above qq-plot, we can see that points almost fall on 45^o line so it means that the distribution of the error terms is normal with mean 0 and constant variance.

Distribution of Response variable:

The best way to check the distribution of any variable is by constructing the histogram.



From the above histogram plot, we can see that the shape of the distribution is bell-shaped or symmetrical which indicates that the distribution of the response variable is approximately normal. From the histogram we can also conclude that the most of students in our dataset have average grade between 10 to 12 which is very common.

Goodness of Fit:

From our model results, $R^{\wedge 2}$ was found to be 0.3345 which means that only 33.45% of the variation is explained by the predictors in our response variable average grade and the remaining 66.55% of the variation is still unexplained which is due to the significant factors that are not included in the model, random variation or measurement error. The model explains only one-third of the variation which indicates that our multiple linear regression model is performing poor. As I observed, there were so many predictors that were not individually statistically significant in the model. However, the model was found statistically significant overall at 5% significance level (p-value < 0.05 and F-statistic = 10.14) which means that there is statistical evidence that at least one of our predictors is statistically significant or in other words our set of predictors is statistically significant when considered together.

Therefore, the final regression model is useful for explaining and predicting students' academic performance, although additional variables could potentially improve its explanatory power.

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